

Analysis and simulation of a renewal process in reliability theory

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An electronic system requires m components to operate. These components are identical and operate independently. The lifespan of each component follows an exponential distribution of mean $1/\lambda$. The system is operational only if there are at least m components operating simultaneously. When the system is not operational, the components that have not failed continue to function. We increase the system availability by placing redundant units and by periodic maintenance. We place r redundant components in addition to the m needed. Through periodic maintenance, the system is inspected every T units of time. At each inspection time, non-operational components are replaced with new ones. The time it takes a technician to replace a component is negligible. Defective units can only be detected in these inspection times. For each working component, there is an operating cost I per unit of time. There is also a fixed cost K for each inspection and a repair cost R for each component that must be replaced.

We will find:

1. *The probability that the system will fail between two consecutive inspection periods.*
2. *The number r of redundant units that would guarantee a probability that the system is operational p_0 .*
3. *The asymptotic cost per unit of time of the system.*
4. *The asymptotic proportion of time that the system is operational.*

General considerations

Let X_i be the lifespan of component i . We assume that $X_i \sim \text{Exp}(\lambda)$. The system is operational when m components work. In addition, we have r redundant components. In total, there are $m + r$ components.

The exponential distribution is memoryless (*the past has no bearing on its future behavior*), so components that are still operational at the time of an inspection continue to operate under the same statistical conditions as components that have been recently renewed. The system starts as new at $t = 0$ and is renewed to the same initial state at times $T, 2T, 3T, \dots$. Therefore, from a probabilistic point of view, it is equivalent to observe the system in the interval $[0, T]$, in $[T, 2T]$, in $[2T, 3T]$, etc. As far as analytical calculations allow, we will obtain explicit expressions according to the 4 variables of the system:

- The minimum number of operational components that the system needs to function (m)
- The number of redundant components (r)
- The failure rate for the components, which is the parameter of the exponential distribution for their lifespans (λ)
- The time between two consecutive inspections (T)

Then, all theoretical calculations will be validated by simulations in **R** which will provide compatible results with high accuracy.

1. The probability that the system will fail between two consecutive inspection periods

The probability that a component is operational in the interval $[0, T]$ is

$$P(X_i > T) = \int_T^\infty \lambda e^{-\lambda t} dt = e^{-\lambda T}$$

We define the random variable

$$Y = \text{number of operational components in } [0, T]$$

A component can only be either operational or non-operational. Among the $m + r$ components, one component is operational with probability $e^{-\lambda T}$. Then,

$$Y \sim \text{Binomial}(m + r, e^{-\lambda T})$$

The system fails when there are less than m operational components. Thus, the probability that the system will fail between two consecutive inspections is

$$P(Y < m) = \sum_{k=0}^{m-1} \binom{m+r}{k} (e^{-\lambda T})^k (1 - e^{-\lambda T})^{m+r-k}$$

In the attached code, this probability is shown for $m = 10$, $r = 12$, $\lambda = 0.25$, $T = 2$. In that case, the probability of system failure is about 0.048, verified both by simulations and by direct application of the above expression.

2. The number r of redundant units that would guarantee a probability that the system is operational p_0 .

The probability that the system is operational is the complementary of that of the previous section and we want it to be at least p_0 .

$$1 - \sum_{k=0}^{m-1} \binom{m+r}{k} (e^{-\lambda T})^k (1 - e^{-\lambda T})^{m+r-k} \geq p_0$$

For a fixed value of m , the above probabilities are calculated for several values of r starting from zero. The first one that exceeds the threshold p_0 is the minimum value of r that guarantees the desired quality.

In the attached code, this probability is shown for $m = 10$, $p_0 = 0.95$, $\lambda = 0.25$, $T = 2$. The probabilities that the system is operational are calculated for $r \in [0, 20]$ forming an increasing curve. The first probability crossing the threshold $p_0 = 0.95$ is approximately 0.952 and corresponds to $r = 12$. The procedure has been carried out by simulating the probabilities and by applying the theoretical formula, obtaining similar results.

3. The asymptotic cost per unit of time of the system.

We wish to compute

$$\lim_{t \rightarrow \infty} \frac{c_{[0,t]}}{t}$$

where $c_{[0,t]}$ is the cost in the interval $[0, t]$.

By the properties of renewal processes, we turn this limit into the study of the expected values in the first interval, $[0, T]$.

$$\lim_{t \rightarrow \infty} \frac{c_{[0,t]}}{t} = \frac{E[c_{[0,T]}]}{E[T]}$$

where $E[T] = T$ because the inspection times are constant.

We compute the random variable $c_{[0,T]}$. The cost is due to:

- *Operating cost:* The operational components cost I monetary units per unit of time. Therefore, we consider the sum of the products of this rate times the lifespan of each component, up to a maximum of T units of time.
- *Inspection cost:* It is a fixed cost K that we add at the end of the interval $[0, T]$.
- *Repair cost:* During the inspection at the end of the interval $[0, T]$, non-operational components ($X_i < T$) add a cost R , whereas operational components ($X_i > T$) keep working at no additional cost.

Therefore,

$$c_{[0,T]} = I \sum_{i=1}^{m+r} \min(X_i, T) + K + R \sum_{i=1}^{m+r} 1_{\{X_i < T\}}$$

where $1_{\{X_i < T\}}$ denotes the indicator function that takes the value 1 when $X_i < T$ and the value 0 when $X_i > T$.

The expected value of the above expression (applying the linearity of the operator) is

$$\begin{aligned} E[c_{[0,T]}] &= E \left[I \sum_{i=1}^{m+r} \min(X_i, T) \right] + E[K] + E \left[R \sum_{i=1}^{m+r} 1_{\{X_i < T\}} \right] = \\ &= I \sum_{i=1}^{m+r} E[\min(X_i, T)] + K + R \sum_{i=1}^{m+r} E[1_{\{X_i < T\}}] \end{aligned}$$

On the one hand, we have

$$\begin{aligned}
E[\min(X_i, T)] &= \int_0^\infty P(\min(X_i, T) > t) dt = \int_0^\infty P(X_i > t)P(T > t) dt = \\
&= \int_0^\infty P(X_i > t) \cdot 1_{\{T > t\}} dt = \int_0^T P(X_i > t) dt = \int_0^T e^{-\lambda t} dt = \frac{1}{\lambda} (1 - e^{-\lambda T})
\end{aligned}$$

On the other hand, we have

$$E[1_{\{X_i < T\}}] = P(X_i < T) = 1 - e^{-\lambda T}$$

We substitute and obtain $E[c_{[0,T]}]$

$$\begin{aligned}
E[c_{[0,T]}] &= I \sum_{i=1}^{m+r} \frac{1}{\lambda} (1 - e^{-\lambda T}) + K + R \sum_{i=1}^{m+r} (1 - e^{-\lambda T}) = \\
&= \frac{I}{\lambda} (m+r) (1 - e^{-\lambda T}) + K + R(m+r) (1 - e^{-\lambda T}) = \\
&= \left(\frac{I}{\lambda} + R \right) (m+r) (1 - e^{-\lambda T}) + K
\end{aligned}$$

Thus, the asymptotic cost per unit of time is

$$\frac{E[c_{[0,T]}]}{E[T]} = \frac{\left(\frac{I}{\lambda} + R \right) (m+r) (1 - e^{-\lambda T}) + K}{T}$$

In the attached code, this cost per unit of time is shown for $m = 10$, $r = 12$, $\lambda = 0.25$, $T = 2$, $I = 1$, $K = 2$, $R = 3$. We ran simulations for 1000 inspection periods to simulate the accumulated cost per unit of time in the long run. Both the simulations and direct application of the above formula provide a value close to 31.3 monetary units per unit of time.

4. The asymptotic proportion of time that the system is operational

Based on the same principle of renewal processes that we used to calculate the asymptotic cost per unit of time, we can compute the asymptotic availability as the quotient of the expected value of the system operation time (U) and the expected value of the total time of the first interval (T).

We order the operation time of the components. The system is operational as long as at least m components are operational. Then, when the component with the m -th largest lifetime breaks down, the system stops working. With the ordering from shortest to longest lifetime, the component with the m -th largest lifetime is the component with the $(r + 1)$ -th smallest lifetime.

The lifetime of the component i in the interval $[0, T]$ is $Z_i = \min(X_i, T)$. Let $Z_{(1)}, \dots, Z_{(m+r)}$ be the ordered sample of lifetimes from shortest to longest. Then, $U = Z_{(r+1)}$. By analyzing the order statistics, it can be shown that, if one has a sample $Z_1, \dots, Z_n$ coming from a distribution with probability density function $f_Z(x)$ and cumulative distribution function $F_Z(x)$, then the PDF of $Z_{(k)}$ (k -th smallest value) is

$$f_{Z_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F_Z(x)]^{k-1} [1 - F_Z(x)]^{n-k} f_Z(x)$$

In our case, $n = m + r$ and we are interested in $k = r + 1$.

The variables Z_i follow a distribution similar to an exponential distribution in the interval $(0, T)$, but the remaining probability mass in (T, ∞) will now be concentrated at T . The probability mass in $(0, T)$ is $\int_0^T \lambda e^{-\lambda t} dt = 1 - e^{-\lambda T}$. Therefore, in T , there must be a probability mass $e^{-\lambda T}$. To model this while ensuring that the probability density is zero in (T, ∞) , we use a Dirac delta centered at T . Thus, the probability density function of $Z_i = \min(X_i, T)$ is

$$f_Z(x) = \begin{cases} 0, & x < 0 \\ \lambda e^{-\lambda x}, & 0 < x < T \\ e^{-\lambda T} \delta_T(x), & x > T \end{cases}$$

Two important properties of a Dirac delta:

$$\int_{-\infty}^{\infty} \delta_T(x) dx = 1, \quad \int_{-\infty}^{\infty} x \delta_T(x) dx = T$$

It follows that the function $f_Z(x)$ is indeed a probability density function since

$$\int_0^{\infty} f_Z(x) dx = \int_0^T \lambda e^{-\lambda x} dx + \int_T^{\infty} e^{-\lambda T} \delta_T(x) dx = 1 - e^{-\lambda T} + e^{-\lambda T} = 1$$

and also,

$$E[Z_i] = \int_0^{\infty} x f_Z(x) dx = \int_0^T x \lambda e^{-\lambda x} dx + \int_T^{\infty} x e^{-\lambda T} \delta_T(x) dx =$$

$$= \frac{1 - (\lambda T + 1) e^{-\lambda T}}{\lambda} + T e^{-\lambda T} = \frac{1}{\lambda} (1 - e^{-\lambda T})$$

which is consistent with the calculation of $E[\min(X_i, T)]$ in the previous section. From the probability density function, we obtain the cumulative distribution function:

$$F_Z(x) = \int_{-\infty}^x f_Z(x) dx = \begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & 0 < x < T \\ 1, & x > T \end{cases}$$

Therefore, the probability density function of $Z_{(r+1)}$ (the $(r+1)$ -th shortest lifetime) is

$$f_{Z_{(r+1)}}(x) = \frac{(m+r)!}{r!(m-1)!} [F_Z(x)]^r [1 - F_Z(x)]^{m-1} f_Z(x)$$

The pieces of this function for $x < T$ are clear because they are analogous to those obtained by substituting the PDF and CDF of an exponential distribution:

$$0, \quad x < 0$$

$$\frac{(m+r)!}{r!(m-1)!} [1 - e^{-\lambda x}]^r [e^{-\lambda x}]^{m-1} \lambda e^{-\lambda x}, \quad 0 < x < T$$

Nevertheless, since the probabilities must add up to 1, we have to proceed in the same way as when we obtained $f_Z(x)$ through a Dirac delta. We integrate the above expression in $(0, T)$ and its complementary will be the remaining probability mass which will be multiplied by a Dirac delta for $x > T$. However, it is difficult to obtain a general closed expression of the integral for any value of m and r . It is possible to define the general probability density function in terms of the incomplete beta function. Nevertheless, the formula for the subsequent calculation of the expected value of $Z_{(r+1)}$ becomes impractical and potentially requires the use of other special functions. Since we want to obtain explicit solutions, we will solve this question for particular values of m and r (two complete examples of the steps will be shown, but the procedure is analogous for any specific values of m and r). This will allow us to solve the integrals easily. On the other hand, working with general values of λ and T does not pose any additional difficulty and we will keep them until the end.

Consider $m = 1$ and $r = 0$. Then, we compute

$$\int_0^T \frac{(1+0)!}{0!(1-1)!} (1 - e^{-\lambda x})^0 (e^{-\lambda x})^{1-1} \lambda e^{-\lambda x} dx = \int_0^T \lambda e^{-\lambda x} dx = 1 - e^{-\lambda T}$$

and define the following PDF (valid for the particular case $m = 1, r = 0$)

$$f_{Z_{(1)}}(x) = \begin{cases} 0, & x < 0 \\ \lambda e^{-\lambda x}, & 0 < x < T \\ e^{-\lambda T} \delta_T(x), & x > T \end{cases}$$

This is the same function as $f_Z(x)$ obtained above, so the expected value is also the same:

$$E[Z_{(1)}] = \frac{1}{\lambda}(1 - e^{-\lambda T})$$

Then, the asymptotic availability is

$$\frac{E[Z_{(1)}]}{E[T]} = \frac{\frac{1-e^{-\lambda T}}{\lambda}}{T} = \frac{1 - e^{-\lambda T}}{\lambda T}$$

Let us consider a more complex case: $m = 3$, $r = 2$. The piece of the PDF of $Z_{(3)}$ in $(0, T)$ is

$$\frac{(3+2)!}{2!(3-1)!} (1 - e^{-\lambda x})^2 (e^{-\lambda x})^{3-1} \lambda e^{-\lambda x} = 30(1 - e^{-\lambda x})^2 (e^{-\lambda x})^2 \lambda e^{-\lambda x}$$

so the following probability mass is accumulated in $(0, T)$:

$$\int_0^T 30(1 - e^{-\lambda x})^2 (e^{-\lambda x})^2 \lambda e^{-\lambda x} dx = 1 - 10e^{-3\lambda T} + 15e^{-4\lambda T} - 6e^{-5\lambda T}$$

In order to obtain the piece of the PDF of $Z_{(3)}$ in (T, ∞) , we consider the complementary probability:

$$1 - (1 - 10e^{-3\lambda T} + 15e^{-4\lambda T} - 6e^{-5\lambda T}) = 10e^{-3\lambda T} - 15e^{-4\lambda T} + 6e^{-5\lambda T}$$

and define the following PDF (valid for the particular case $m = 3$, $r = 2$)

$$f_{Z_{(3)}}(x) = \begin{cases} 0, & x < 0 \\ 30(1 - e^{-\lambda x})^2 (e^{-\lambda x})^2 \lambda e^{-\lambda x}, & 0 < x < T \\ (10e^{-3\lambda T} - 15e^{-4\lambda T} + 6e^{-5\lambda T}) \delta_T(x), & x > T \end{cases}$$

The expected value of $Z_{(3)}$ is

$$\begin{aligned} E[Z_{(3)}] &= \int_0^T x \cdot 30(1 - e^{-\lambda x})^2 (e^{-\lambda x})^2 \lambda e^{-\lambda x} dx + \int_T^\infty x (10e^{-3\lambda T} - 15e^{-4\lambda T} + 6e^{-5\lambda T}) \delta_T(x) dx = \\ &= \frac{47 - 200(3\lambda T + 1)e^{-3\lambda T} + 225(4\lambda T + 1)e^{-4\lambda T} - 72(5\lambda T + 1)e^{-5\lambda T}}{60\lambda} + (10e^{-3\lambda T} - 15e^{-4\lambda T} + 6e^{-5\lambda T})T = \\ &= \frac{47 - 200e^{-3\lambda T} + 225e^{-4\lambda T} - 72e^{-5\lambda T}}{60\lambda} \end{aligned}$$

Then, the asymptotic availability is

$$\frac{E[Z_{(3)}]}{E[T]} = \frac{47 - 200e^{-3\lambda T} + 225e^{-4\lambda T} - 72e^{-5\lambda T}}{60\lambda T}$$

In the attached code, this asymptotic availability is shown for $m = 3$, $r = 2$, $\lambda = 0.25$, $T = 2$. We ran simulations for 1000 inspection periods to simulate how much time the system is operational in the long run. Both the simulations and direct application of the above formula provide a value close to 0.897, indicating that, in that case, the system tends to be operational 89.7% of the time.